

An Introduction to Applied Bayesian Statistics with Stan

2026-08-17



Overview

Course Goals

- Provide the foundation and orientation so that you can
 - Use Stan (or **brms**) in your work
 - Understand and enjoy StanCon talks!

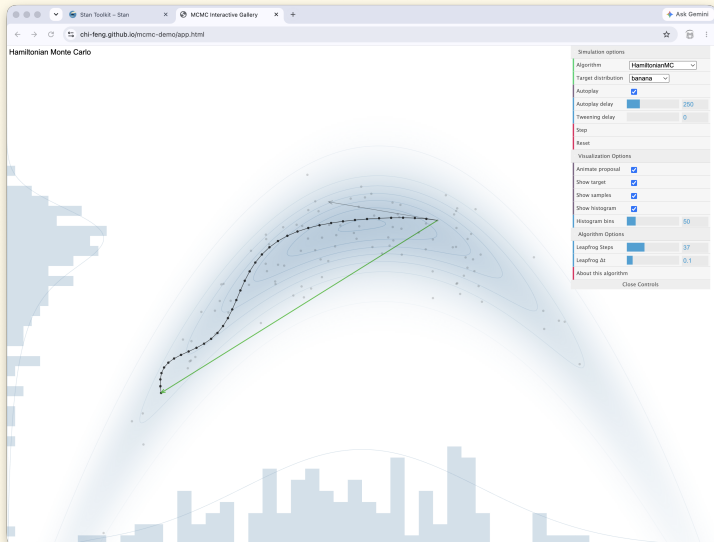
Outline

- About Stan ✓
- Bayesian models and workflow ✓
- Model specification in Stan ✓
- Multilevel model specification in **brms** ✓
- HMC and multi-level models ←

Hamiltonian Monte Carlo

Hamiltonian Monte-Carlo

MCMC Interactive Gallery - <https://chi-feng.github.io/mcmc-demo/app.html>



Hamiltonian Monte-Carlo

- Generate draws from the **joint** log posterior density $\log p(\theta, y)$
 - follow the **Hamiltonian flow** according to the **gradient** of the density
 - Stan's automatic-differentiation computes these gradients
- HMC computes on the unconstrained scale $\mathbb{R}^N$
 - Stan provides constraining/unconstraining transforms
- The Hamiltonian flow is continuous; discretized to **steps**
 - One draw from the posterior consists of many steps
- HMC parameters must be tuned for the posterior geometry
 - Tune step size and estimate mass matrix during warmup, determine number of steps on-the-fly (NUTS)
- See Michael Betancourt's excellent [A Conceptual Introduction to Hamiltonian Monte Carlo](#)

Euclidean Hamiltonian Monte Carlo

- **Phase space:** q position (parameters); p momentum
- **Target density:** $\pi(q)$
- **Potential energy:** $V(q) = -\log \pi(q)$
- **Metric (mass matrix):** M
 - rescales and rotates the dynamics to match posterior geometry
- **Momentum refresh:** $p \sim \text{MultiNormal}(0, M)$
- **Kinetic energy:** $T(p) = \frac{1}{2} p^\top M^{-1} p$
- **Hamiltonian:** $H(p, q) = V(q) + T(p)$

$$\frac{dq}{dt} = M^{-1} p$$

$$\frac{dp}{dt} = -\nabla_q \log \pi(q)$$

Leapfrog integrator

- Solves Hamilton's equations by **simulating dynamics**
(symplectic [volume preserving]; ϵ^3 error per step, ϵ^2 total error)
- Given: **step size** ϵ , **mass matrix** M , **parameters** q
- **Initialize kinetic** energy, $p \sim \text{Normal}(0, \mathbf{I})$
- **Repeat** for L leapfrog steps:

$$p \leftarrow p - \frac{\epsilon}{2} \frac{\partial V(q)}{\partial q} \quad [\text{half step in momentum}]$$

$$q \leftarrow q + \epsilon M^{-1} p \quad [\text{full step in position}]$$

$$p \leftarrow p - \frac{\epsilon}{2} \frac{\partial V(q)}{\partial q} \quad [\text{half step in momentum}]$$

HMC Tuning Parameters

- HMC simulate Hamiltonian trajectory of a particle through parameter space
 - at each iteration, provide a refresh of random momentum ρ where

$$\rho \sim \text{MultiNormal}(0, M),$$

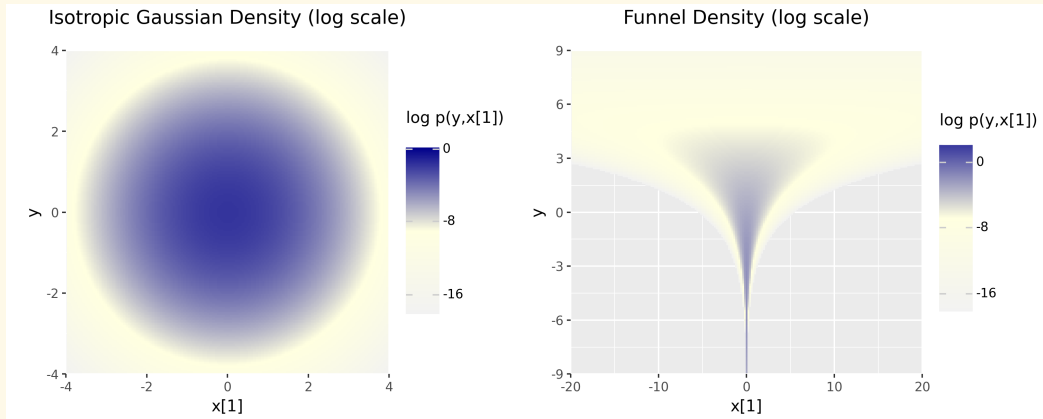
where M is the positive definite mass matrix

- this puts posterior directions on comparable scales
(we have a high-dimensional posterior)
- The well-tuned HMC sampler requires that:
 - one trajectory is composed of just enough steps
 - the step-size just small enough to follow the Hamiltonian
 - the mass matrix is a good *linear pre-conditioner*

Efficiency Tuning

- The HMC sampler explores the posterior density of the *parameter space*
- If the posterior is (approximately) multivariate normal the mass matrix produces an isotropic Gaussian posterior - a single step in every direction is accurate
- If the posterior is sharply curved, no single mass matrix is valid everywhere
 - during warmup the step-size is adjusted smaller and smaller
 - the sampler fails to make progress
- We can rewrite Stan programs in terms of *transformed parameters*
 - see the Stan User's Guide [Efficiency Tuning](#) chapter.
- We have already seen this: the centered and non-centered parameterizations of a hierarchical model

Efficiency Tuning for Hierarchical Models



Easy vs. hard posterior densities

Efficiency Tuning for Hierarchical Models

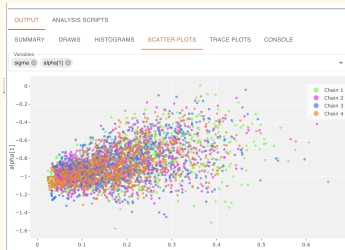
- The isotropic Gaussian density is *not* a hierarchical model
 - it is the product of N normal distributions
- The funnel distribution is hierarchical
 - the scale of elements x depends on global parameter y
- **But** *the posterior density is the product of the prior times the likelihood*
 - shape of the posterior morphs as the model sees more data
- You see this already in the Stan Playground scatter plots

Baseball Example Posterior Densities

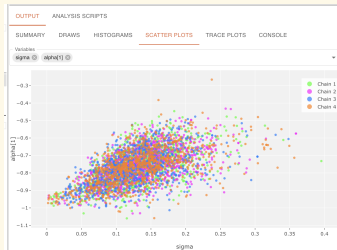
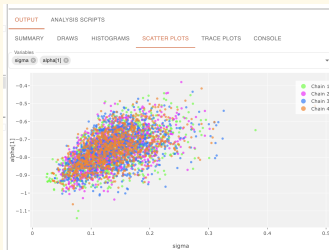
Centered parameterization

Non-centered parameterization

Low

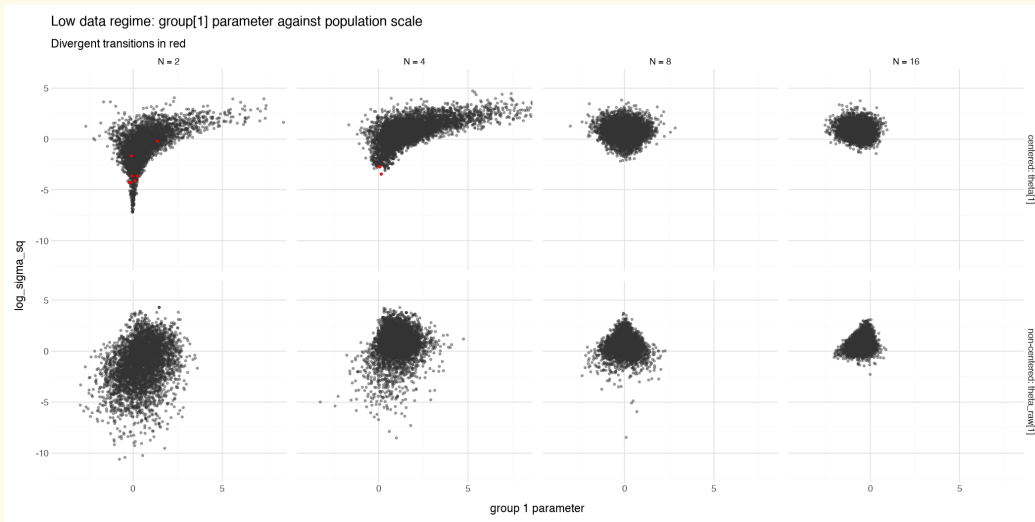


Higher

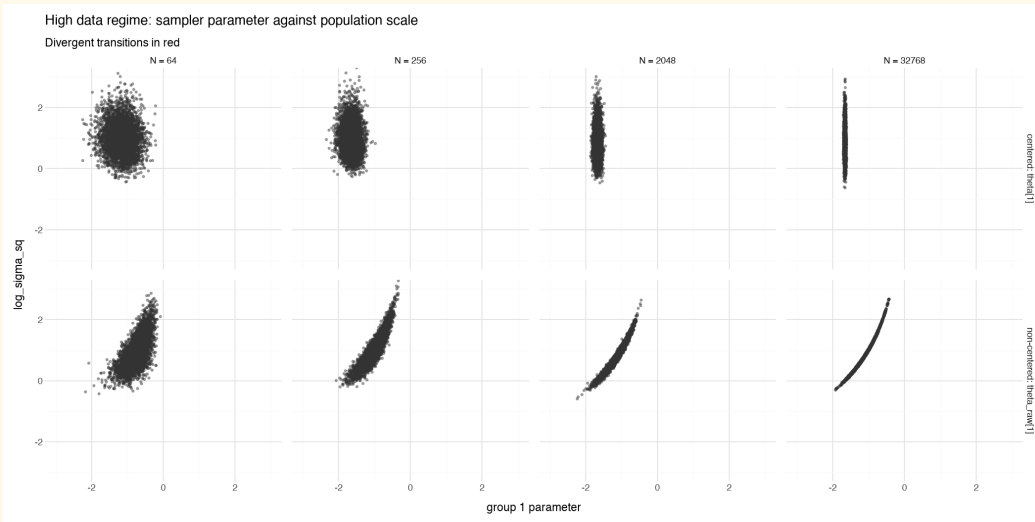


- The R-hat statistic and ESS are related
 - see [Nested R-hat: Assessing the convergence of Markov chain Monte Carlo when running many short chains](#)
 - a further consequence is slow sampling time (low ESS/s) as well.
- In low data regimes the non-centered parameterization improves performance
 - better ESS
 - faster total sampling time
- In high data regimes the centered parameterization wins
- The baseball data is an illustration of “low data” and “more data”

Centered vs. Non-Centered Parameterization: Small Data



Centered vs. Non-Centered Parameterization: Big Data



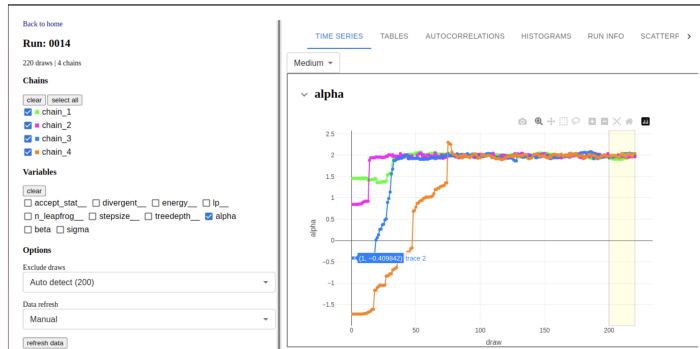
Centered vs. Non-Centered Parameterization: Complicated Data

- These plots were generated from synthetic data generated according to the model
- Beyond a few simple cases, there is no one good solution
- *Note:* **brms** always uses the non-centered parameterization
- Some References
 - [A General Framework for the Parametrization of Hierarchical Models](#)
Papaspiliopoulos et al, 2007
 - [Hamiltonian Monte Carlo for Hierarchical Models](#)
Betancourt and Girolami, 2013
 - The Stan User's Guide chapters:
[Problematic Posteriors](#), [Reparameterizations](#) [Efficiency Tuning](#)

Diagnosing Complications: the MCMC monitor

MCMC Monitor : <https://github.com/flatironinstitute/mcmc-monitor>

MCMC Monitor enables tracking and visualization of MCMC processes executed with [Stan](#) in local or remote web browsers. When you run a sampler, you can configure Stan to generate output to a directory on your computer. MCMC Monitor reads this output and displays it in the web app, with real-time updates. As you track the progress of the run, MCMC provides diagnostic plots and statistics.



Questions and Discussion

Stan Documentation

- Stan Documentation Suite
 - [Stan User's Guide](#)
 - [Reference Manual](#) - just the facts
 - [Functions Reference](#) - Stan math library functions available in the Stan language
- Learning Resources
 - [Tutorials](#) - just a few of many
 - [Stan Case Studies](#) - best practices, deep dives for many models
 - [StanCon Talks](#) - past StanCons
 - [Field Guides](#) - education, ecology, epidemiology, cogsci, pharma
- [Stan Forums](#) - your questions answered